

Final Report

Better Place Drones

Propulsion

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Team Members:

Sunit Bapat

Parker Bredice

Kenny Dinh

Michael Hakim

John Kelley

Abdullah Mohammed Bagudu

Sahaj Patel

Finn Wolff

Georgia Institute of Technology

Introduction

As the propulsion subsystem team for the XPRIZE Wildfire Challenge project, our responsibility was to design, integrate, and optimize the engine powering the autonomous wildfire-response drone. This included developing efficient cooling methods, determining proper mounting solutions for stability and vibration control, and conducting systematic testing to compare configurations, accessories, and performance characteristics. Through these efforts, we aimed to ensure that the propulsion system remained reliable, efficient, and structurally compatible with the overall drone design.

The objective of the overall team was to create a robust and mission-capable engine system that enables the drone to autonomously locate, travel to, and extinguish wildfires within a designated area, reload suppression material, and repeat the cycle. Achieving this requires balancing high-power output with thermal regulation, endurance, weight limitations, and long-term reliability. Ultimately, our goal as the propulsion sub team was to support the development of a propulsion solution that could withstand the operational demands of wildfire environments while meeting the performance standards required by the XPRIZE competition.

Methodology

The main software used was the Intelliject software. This software allows us to control the engine in terms of power output. It also allowed us to view live performance data such as the RPM. Our process for testing the engine was to first check the hardware, adjust the settings on the Intelliject software, and then use a drill to start the engine. We then evaluated the data recorded by the software, made adjustments to the settings, and tried again. We also would occasionally test individual parts of the engine to see whether specific processes were functional.

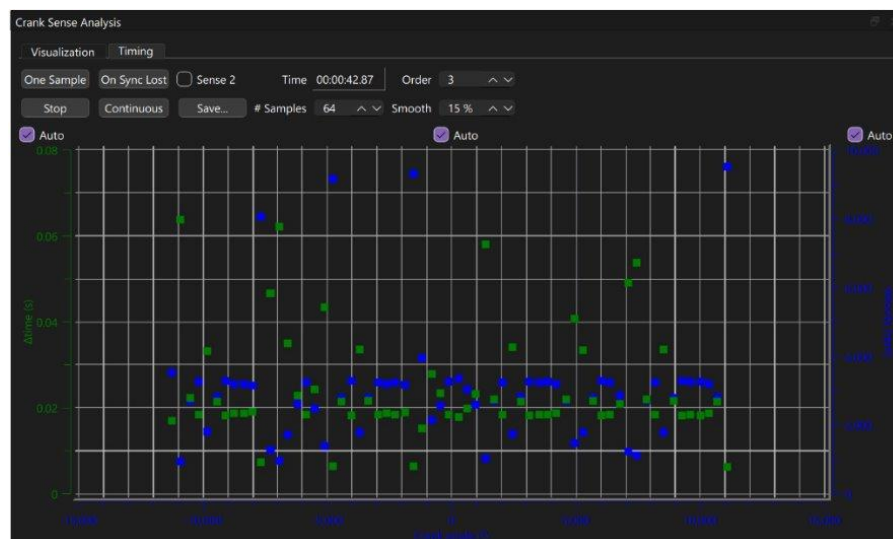
Our process followed a cycle of preparation, testing, evaluation, and refinement. We began each test by checking the hardware to make sure the engine and all components were properly set up. After that, we adjusted the required settings within the Intelliject software to prepare for startup. Each trial ended with an analysis of the data so we could make informed adjustments for the next attempt.

Solution Development/Design & Results

The general issue the propulsion team has been investigating since the beginning of the semester is troubleshooting issues in the engine system and how to eventually implement it into the drone power. In particular, the wiring between Trampa board and hall sensors were adjusted for proper RPM readings as well as corrections and better organization made between wiring in the proper CAN device (Systec). Additionally, the team has been able to narrow down the points of issue through data collection as mentioned previously and confirmation through the manual and outside sources (contact with ECE professors and calls to Cobra Aero). The comparison of the crank sense analysis data can be seen below in Figure 1. It displays proper sensing between

the change in time and angle of the crank sense in the wheel. It was confirmed that the occasional error of “crank sense synchronization” is not preventing the engine starting because it is not categorized as a sticky error in the Intelliject software. As of recently the problem of the engine not firing upon start has been diagnosed exclusively with malfunctions in the ignition box. Since the last team meeting, a new ignition box has been ordered and implemented into the engine system, although additional testing may be needed.

The propulsion team has also been prototyping potential mounting concepts for the engine, as shown in Figures 2, 3, and 4. A vertical mounting orientation is currently the favored configuration due to its cooling implications. This orientation is also lighter and easier to integrate than a horizontal configuration. The team has also researched cooling methods for the engine shown below. Lightweight server cooling fans were considered as they provide higher airflow efficiency and improved cooling performance at a lighter weight compared to other fans. Heat sinks were evaluated but ruled out due to their relatively high weight and due to integration challenges. Shrouds and casings were also considered to channel airflow through the heat sink to improve heat transfer efficiency and overall cooling performance. Additionally, coatings such as Jet Hot and White Lightning Exhaust Coatings have been researched to potentially apply to critical engine locations to enhance convective cooling and provide crucial radiant heat shielding against wildfire exposure. These coatings add minimal mass to the system while providing significant thermal protection benefits.



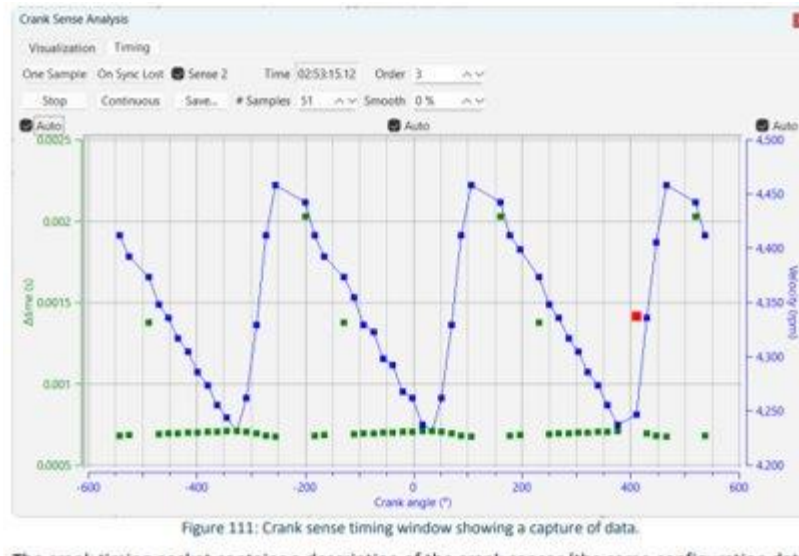


Figure 1: Crank Sense Analysis Data comparison displaying the crank angle as a function of the time delay between successful senses and the velocity (RPM).

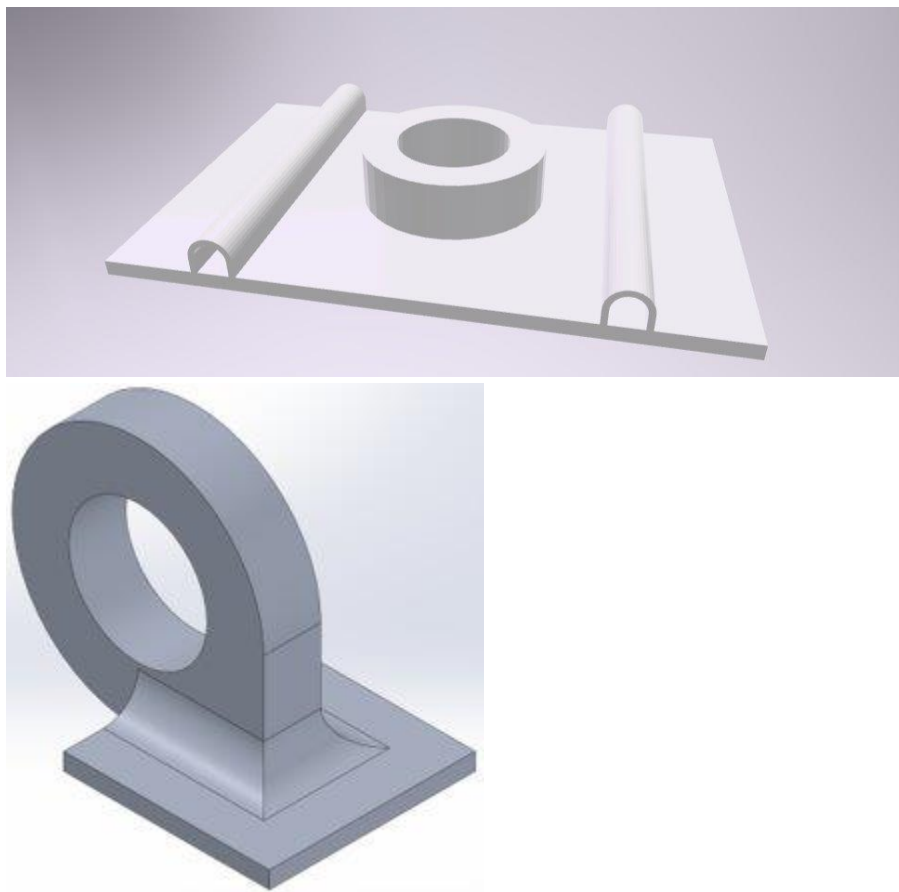


Figure 2: Mounting CAD

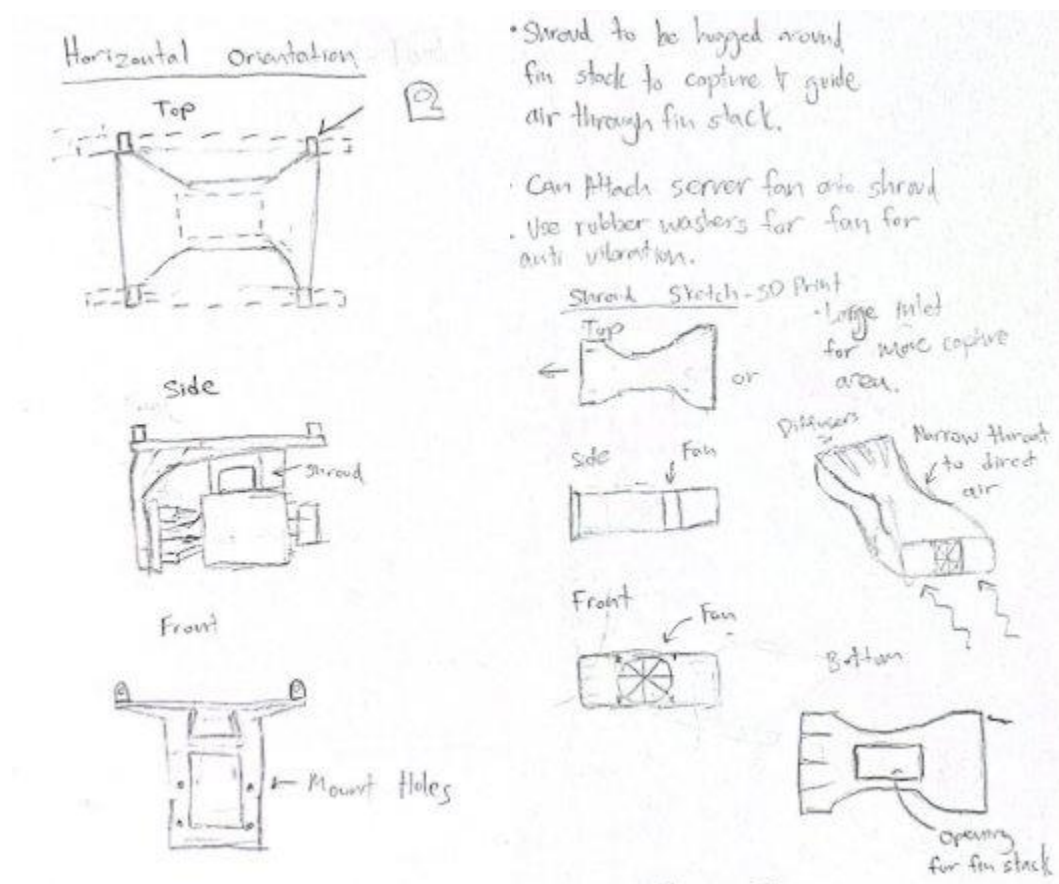


Figure 3: Mounting sketches and concepts

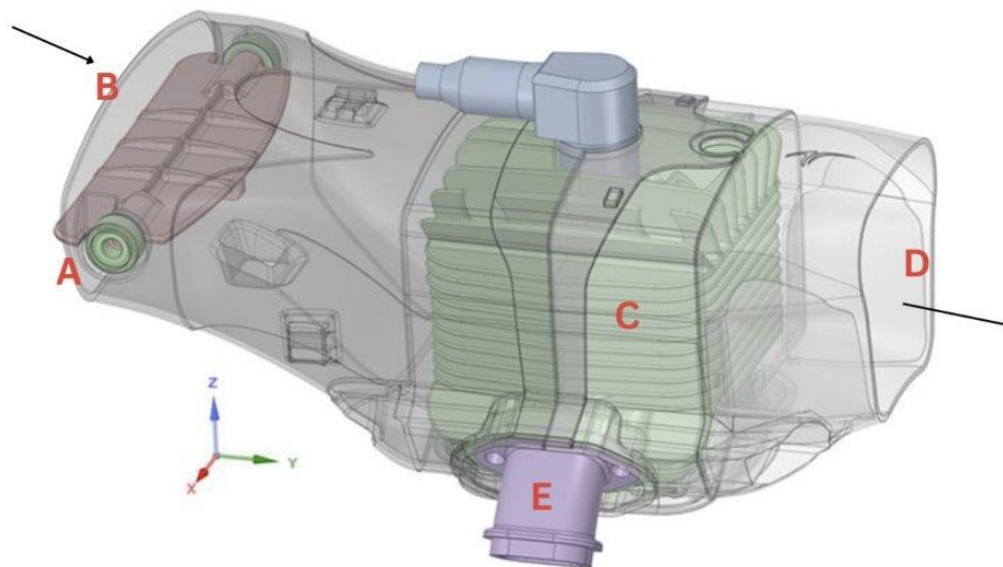


Figure 4: Cooling configuration using shroud/casing

Lessons Learned & Challenges

Throughout the semester, we gained valuable experience not only in technical development but also in collaborative engineering practices. We learned the importance of iterative design, where testing, failure, and redesign were necessary steps in refining our propulsion system. Effective communication across subsystems became essential, as decisions made by the propulsion team directly affected weight distribution, structural integrity, and overall drone performance. We also developed a stronger understanding of the balance between theoretical performance and real-world constraints such as manufacturability, safety, and resource availability.

As a team, we encountered several key challenges during the development process. One major difficulty was the limited communication with Cobra Aero, which slowed our ability to troubleshoot engine issues and obtain technical support when needed. Additionally, repeated engine startup failures significantly delayed testing, requiring extra troubleshooting time before meaningful performance data could be collected. Finally, inconsistent punctuality within the team affected our workflow and made coordination more difficult, especially when tasks depended on timely collaboration across subsystem groups. These challenges highlighted areas for improvement in communication, accountability, and project management moving forward.

Future Work

Moving forward, the propulsion sub team will continue engine testing to refine startup reliability, performance consistency, and operational safety. The data collected from these tests will be analyzed to determine the most efficient engine configuration, including optimal accessories, mounting methods, and cooling strategies. Once a reliable setup is established, the next phase will involve integrating the propulsion system into the larger drone platform and conducting flight tests to evaluate performance under real operating conditions. This progression will help ensure the propulsion system is fully optimized and ready to meet the demands of the XPRIZE wildfire mission.